Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2011 – Paper 2 – Solutions**

June 2011 – Question 1 – Solution

(a) Without using calculators, find the exact value of

(i) $\left(\sqrt{75}+\sqrt{12}\right)^2-\left(\sqrt{75}-\sqrt{12}\right)^2$ [3]

$$\left(\sqrt{75}+\sqrt{12}\right)^2-\left(\sqrt{75}-\sqrt{12}\right)^2$$

$$=\left(\sqrt{75}\right)^2+2\left(\sqrt{75}\right)\left(\sqrt{12}\right)+\left(\sqrt{12}\right)^2-\left[\left(\sqrt{75}\right)^2-2\left(\sqrt{75}\right)\left(\sqrt{12}\right)+\left(\sqrt{12}\right)^2\right]$$

$$=75+2\sqrt{75}\sqrt{12}+12-\left(75-2\sqrt{75}\sqrt{12}+12\right)$$

$$=75+2\sqrt{75}\sqrt{12}+12-75+2\sqrt{75}\sqrt{12}-12$$

$$=4\sqrt{75}\sqrt{12}$$

$$=4\sqrt{25}\sqrt{3}\sqrt{4}\sqrt{3}$$

$$=4(5)\sqrt{3}(2)\sqrt{3}$$

$$=4(5)(2)(3)$$

$$=120$$

(ii) $27^{\frac{1}{4}}\times 9^{\frac{3}{8}}\times 81^{\frac{1}{8}}$ [3]

$$27^{\frac{1}{4}}\times 9^{\frac{3}{8}}\times 81^{\frac{1}{8}}$$

$$=(3^3)^{\frac{1}{4}}\times(3^2)^{\frac{3}{8}}\times(3^4)^{\frac{1}{8}}$$

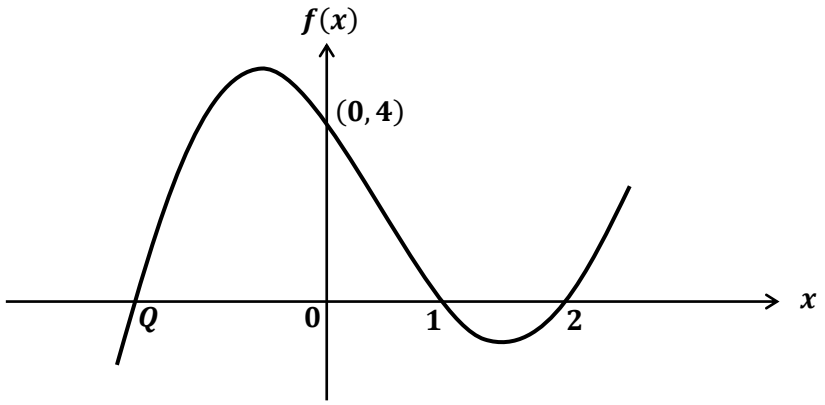
$$=(3)^{\frac{3}{4}}\times(3)^{\frac{3}{4}}\times(3)^{\frac{1}{2}}$$

$$=(3)^{\frac{3}{4}+\frac{3}{4}+\frac{1}{2}}$$

$$=3^2$$

$$=9$$

(b) The diagram below, **not drawn to scale**, represents a segment of the graph of the function $f(x) = x^3 + mx^2 + nx + p$ where m, n and p are constants.



Find

(i) the value of p [2]

$f(x) = x^3 + mx^2 + nx + p$

Substituting the point $(0, 4)$ gives,

$4 = (0)^3 + m(0)^2 + n(0) + p$

$4 = 0 + 0 + 0 + p$

$4 = p$

$\therefore$ The value of $p = 4$.

(ii) the values of m and n [4]

$f(x) = x^3 + mx^2 + nx + 4$

From the graph, points are $(1, 0)$ and $(2, 0)$.

Substituting the point $(1, 0)$ gives,

$$0 = (1)^3 + m(1)^2 + n(1) + 4$$

$$0 = 1 + m + n + 4$$

$$0 = 5 + m + n$$

$-m - 5 = n$ $\rightarrow$ Equation 1

Substituting the point $(2, 0)$ gives,

$0 = (2)^3 + m(2)^2 + n(2) + 4$

$0 = 8 + 4m + 2n + 4$

$0 = 12 + 4m + 2n$

$(\div 2)$

$0 = 6 + 2m + n$ $\rightarrow$ Equation 2

Substituting Equation 1 into Equation 2 gives,

$0 = 6 + 2m + (-m - 5)$

$0 = 6 + 2m - m - 5$

$0 = 1 + m$

$m = -1$

Substituting $m = -1$ into Equation 1 gives,

$n = -(-1) - 5$

$= 1 - 5$

$= -4$

$\therefore$ The values of m and n are $m = -1$ and $n = -4$.

(iii) the x-coordinate of the point $\boldsymbol{Q}$. [2]

$f(x) = x^3 + mx^2 + nx + p$

Since $p = 4, m = -1$ and $n = -4$, then,

$f(x) = x^3 - x^2 - 4x + 4$

Let the coordinates of point Q be $(q, 0)$.

The curve cuts the x-axis at three points $(q, 0), (1, 0)$ and $(2, 0)$.

Hence, the factors of $f(x)$ are $(x - q)$, $(x - 1)$ and $(x - 2)$.

Now,

$f(x) = (x - q)(x - 1)(x - 2)$

So, we have,

$x^3 - x^2 - 4x + 4 = (x - q)(x - 1)(x - 2)$

Equating the constants gives:

$4 = (-q)(-1)(-2)$

$4 = -2q$

$\frac{4}{-2} = q$

$-2 = q$

So, the coordinates of Q are $(-2, 0)$.

$\therefore$ The x-coordinate of the point Q is -2.

(c) (i) By substituting $y = \log_2 x$, or otherwise, solve, for x, the equation

$$\sqrt{\log_2 x} = \log_2 \sqrt{x}$$ [6]

$\sqrt{\log_2 x} = \log_2 \sqrt{x}$

$(\log_2 x)^{\frac{1}{2}} = \log_2 x^{\frac{1}{2}}$

$(\log_2 x)^{\frac{1}{2}} = \frac{1}{2}\log_2 x$

Let $y = \log_2 x$.

$y^{\frac{1}{2}} = \frac{1}{2}y$

Squaring both sides gives:

$\left(y^{\frac{1}{2}}\right)^2 = \left(\frac{1}{2}y\right)^2$

$y = \frac{1}{4}y^2$

$(\times 4)$

$4y = y^2$

$y^2 - 4y = 0$

$y(y-4) = 0$

Either $y = 0$ or $y - 4 = 0$

$y = 4$

When $y = 0$,

$\log_2 x = 0$

$x = 2^0$

$x = 1$

When $y = 4$

$\log_2 x = 4$

$x = 2^4$

$x = 16$

$\therefore x = 1$ and $x = 16$

(ii) Solve, for real value of x, the inequality

$$x^2 - |x| - 12 < 0 \qquad [5]$$

By definition,

$$|x| = \begin{cases} -x & , x < 0 \\ x & , x \geq 0 \end{cases}$$

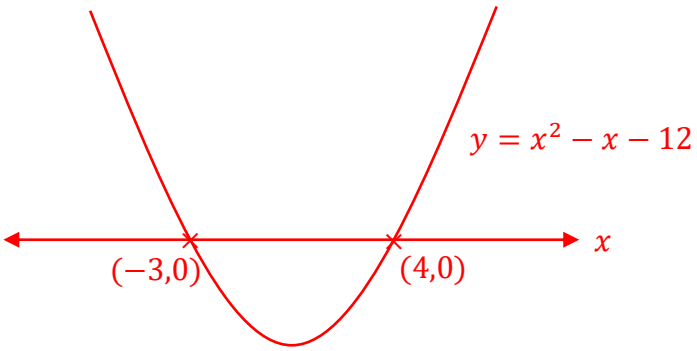
<u>Case 1:</u>

$x^2 - x - 12 < 0$

$(x + 3)(x - 4) < 0$

The critical values are $x = -3$ and $x = 4$.

The sketch of the graph of $y = x^2 - x - 12$ is shown below.



Since we are solving $x^2 - x - 12 < 0$, we look where the graph is negative.

The graph is negative for $-3 < x < 4$.

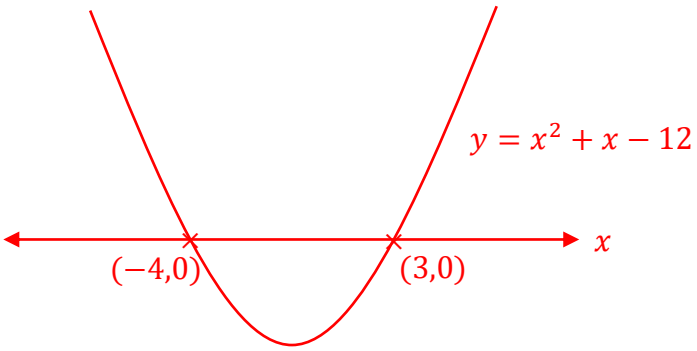
<u>Case 2:</u>

$x^2 + x - 12 < 0$

$(x + 4)(x - 3) < 0$

The critical values are $x = -4$ and $x = 3$.

The sketch of the graph of $y = x^2 + x - 12$ is shown below.

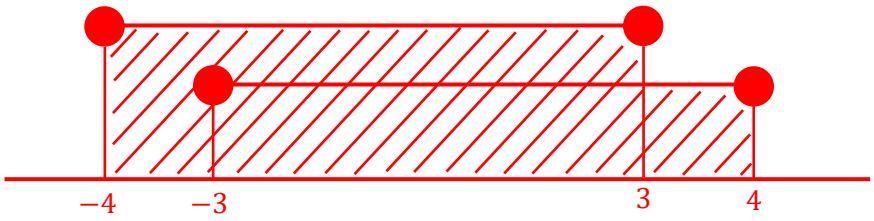


Since we are solving $x^2 + x - 12 < 0$, we look where the graph is negative.

The graph is negative for $-4 < x < 3$.

The solution set is $-3 < x < 4$ or $-4 < x < 3$.

Now, consider the diagram below.



$\therefore$ The solution is $\{x: -4 < x < 4\}$.

Total: 25 marks

June 2011 – Question 2 – Solution

(a) The quadratic equation $x^2 - px + 24 = 0, p \in \mathbb{R}$, has roots α and β.

(i) Express in terms of p

(a) $\alpha + \beta$ [1]

$x^2 - px + 24 = 0$ which is in the form $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

Comparing coefficients of x, we get:

$\alpha + \beta = p$

(b) $\alpha^2 + \beta^2$ [4]

$x^2 - px + 24 = 0$ which is in the form $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

Comparing constants, we get:

$\alpha\beta = 24$

$(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$

Rearranging gives:

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Substituting $\alpha + \beta = p$ and $\alpha\beta = 24$ gives:

$\alpha^2 + \beta^2 = p^2 - 2(24)$

$= p^2 - 48$

(ii) Given that $\alpha^2 + \beta^2 = 33$, find the possible values of p. [3]

Since $\alpha^2 + \beta^2 = p^2 - 48$ and $\alpha^2 + \beta^2 = 33$, then

$p^2 - 48 = 33$

$p^2 = 33 + 48$

$p^2 = 81$

$p = \sqrt{81}$

$p = \pm 9$

$\therefore$ The possible values of p are $p = -9$ and $p = 9$.

(b) The function $f(x)$ has the property that

$$f(2x + 3) = 2f(x) + 3, \quad x \in \mathbb{R}$$

If $f(0) = 6$, find the value of

(i) $f(3)$ [4]

$f(2x + 3) = 2f(x) + 3$

Let $x = 0$.

$f[2(0) + 3] = 2f(0) + 3$

$f(3) = 2f(0) + 3$

$= 2(6) + 3$

$= 12 + 3$

$= 15$

$\therefore f(3) = 15$

(ii) $f(9)$ [2]

Consider,

$2x + 3 = 9$

$2x = 9 - 3$

$2x = 6$

$x = \frac{6}{2}$

$x = 3$

$f(2x + 3) = 2f(x) + 3$

Let $x = 3$.

$f[2(3) + 3] = 2f(3) + 3$

$f(9) = 2f(3) + 3$

$= 2(15) + 3$

$= 30 + 3$

$= 33$

(iii) $f(-3)$ [3]

Consider,

$2x + 3 = -3$

$2x = -3 - 3$

$2x = -6$

$x = \frac{-6}{2}$

$x = -3$

$f(2x+3) = 2f(x) + 3$

Let $x = -3$.

$f[2(-3)+3] = 2f(-3) + 3$

$f(-3) = 2f(-3) + 3$

$-3 = 2f(-3) - f(-3)$

$-3 = f(-3)$

$\therefore f(-3) = -3$

(c) Prove that the product of any two consecutive integers k and $k+1$ is an even integer. [2]

Let k and $k+1$ be two consecutive integers.

Consider k to be even.

Then, $k = 2n, n \in \mathbb{Z}$.

Also, $k+1 = 2n+1, n \in \mathbb{Z}$.

Now,

$k(k+1) = (2n)(2n+1)$

$= 4n^2 + 2n$

$= 2(2n^2 + n)$ which is divisible by 2.

Hence, $k(k+1)$ is even.

Now, consider k to be odd.

Then, $k = 2n - 1, n \in \mathbb{Z}$.

Also, $k + 1 = 2n, n \in \mathbb{Z}$.

Now,

$k(k+1) = (2n-1)(2n)$

$= 4n^2 - 2n$

$= 2(2n^2 - n)$ which is divisible by 2.

Hence, $k(k+1)$ is even.

$\therefore$ The product of any two consecutive integers k and $k+1$ is an even integer.

Q.E.D.

(d) Prove, by mathematical induction, that $n(n^2 + 5)$ is divisible by 6 for all positive integers n. [6]

Let $P(n)$ be the statement:

$n(n^2 + 5)$ is divisible by 6 for $n \in \mathbb{N}$

Looking at $P(1)$.

$(1)(1^2 + 5) = 1 + 5$

$= 6$ which is divisible by 6.

$\therefore P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $k(k^2+5)$ is divisible by 6

$\therefore k(k^2+5) = 6m$, where $m \in \mathbb{N}$

$\Rightarrow k^3 + 5k = 6m$

Looking at $P(k+1)$.

That is, $(k+1)[(k+1)^2+5]$ is divisible by 6.

We need to prove $P(k+1)$.

$$
\begin{aligned}
(k+1)[(k+1)^2+5] &= (k+1)(k^2+2k+1+5) \\
&= (k+1)(k^2+2k+6) \\
&= k^3+2k^2+6k+k^2+2k+6 \\
&= k^3+3k^2+8k+6 \\
&= k^3+3k^2+5k+3k+6 \\
&= (k^3+5k)+3k^2+3k+6 \\
&= 6m+3k^2+3k+6 \qquad [\because \text{assumption}]
\end{aligned}
$$

We now need to prove that $3k^2+3k+6$ is divisible by 6.

Let $P(r)$ be the statement:

$3r^2+3r+6$ is divisible by 6 for $k \in \mathbb{N}$

Assume $P(r)$ is true.

That is, $3r^2 + 3r + 6$ is divisible by 6.

$\therefore 3r^2 + 3r + 6 = 6p$, where $p \in \mathbb{N}$

$\Rightarrow 3r^2 + 3r = 6p - 6$

Looking at $P(r+1)$.

That is, $3(r+1)^2 + 3(r+1) + 6$ is divisible by 6.

We need to prove $P(r+1)$.

$$\begin{aligned}
3(r+1)^2 + 3(r+1) + 6 &= 3(r^2 + 2r + 1) + 3r + 3 + 6 \\
&= 3r^2 + 6r + 3 + 3r + 3 + 6 \\
&= 3r^2 + 9r + 12 \\
&= (3r^2 + 3r) + 6r + 12 \\
&= 6p - 6 + 6r + 12 \\
&= 6p + 6r + 6 \\
&= 6(p + r + 1) \quad \text{which is divisible by 6.}
\end{aligned}$$

Hence, $(k+1)[(k+1)^2 + 5] = 6m + 3k^2 + 3k + 6$ is divisible by 6.

$\therefore$ If $P(k)$ is true, $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$n(n^2 + 5)$ is divisible by 6 for $n \in \mathbb{N}$.

Q.E.D.

Total: 25 marks

June 2011 – Question 3 – Solution

(a)(i) Let $\boldsymbol{a} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}}$ and $\boldsymbol{b} = b_1\hat{\boldsymbol{\imath}} + b_2\hat{\boldsymbol{\jmath}}$ and $|\boldsymbol{a}| = 13$ and $|\boldsymbol{b}| = 10$. Find the value of

$(\boldsymbol{a} + \boldsymbol{b}) \cdot (\boldsymbol{a} - \boldsymbol{b})$. [5]

Consider,

$$\boldsymbol{a} + \boldsymbol{b} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}} + b_1\hat{\boldsymbol{\imath}} + b_2\hat{\boldsymbol{\jmath}}$$

$$= (a_1 + b_1)\hat{\boldsymbol{\imath}} + (a_2 + b_2)\hat{\boldsymbol{\jmath}}$$

And,

$$\boldsymbol{a} - \boldsymbol{b} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}} - b_1\hat{\boldsymbol{\imath}} - b_2\hat{\boldsymbol{\jmath}}$$

$$= (a_1 - b_1)\hat{\boldsymbol{\imath}} + (a_2 - b_2)\hat{\boldsymbol{\jmath}}$$

Consider,

$$\boldsymbol{a} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}}$$

$$|\boldsymbol{a}| = \sqrt{(a_1)^2 + (a_2)^2}$$

$$13 = \sqrt{(a_1)^2 + (a_2)^2}$$

$$(a_1)^2 + (a_2)^2 = (13)^2$$

$$(a_1)^2 + (a_2)^2 = 169$$

And,

$$\boldsymbol{b} = b_1\hat{\boldsymbol{\imath}} + b_2\hat{\boldsymbol{\jmath}}$$

$$|\boldsymbol{b}| = \sqrt{(b_1)^2 + (b_2)^2}$$

$$10 = \sqrt{(b_1)^2 + (b_2)^2}$$

$$(b_1)^2 + (b_2)^2 = (10)^2$$

$$(b_1)^2 + (b_2)^2 = 100$$

Now,

$$(\boldsymbol{a}+\boldsymbol{b})\cdot(\boldsymbol{a}-\boldsymbol{b}) = \begin{pmatrix} a_1+b_1 \\ a_2+b_2 \end{pmatrix} \cdot \begin{pmatrix} a_1-b_1 \\ a_2-b_2 \end{pmatrix}$$

$$= (a_1+b_1)(a_1-b_1)+(a_2+b_2)(a_2-b_2)$$

$$= [(a_1)^2-(b_1)^2]+[(a_2)^2-(b_2)^2]$$

$$= [(a_1)^2+(a_2)^2]-[(b_1)^2+(b_2)^2]$$

$$= 169-100$$

$$= 69$$

(ii) If $2\boldsymbol{b}-\boldsymbol{a} = 11\hat{\boldsymbol{\imath}}$, determine the possible values of $\boldsymbol{a}$ and $\boldsymbol{b}$. [5]

$$2\boldsymbol{b}-\boldsymbol{a} = 11\hat{\boldsymbol{\imath}}$$

$$2(b_1\hat{\boldsymbol{\imath}}+b_2\hat{\boldsymbol{\jmath}})-(a_1\hat{\boldsymbol{\imath}}+a_2\hat{\boldsymbol{\jmath}}) = 11\hat{\boldsymbol{\imath}}$$

$$2b_1\hat{\boldsymbol{\imath}}+2b_2\hat{\boldsymbol{\jmath}}-a_1\hat{\boldsymbol{\imath}}-a_2\hat{\boldsymbol{\jmath}} = 11\hat{\boldsymbol{\imath}}$$

$$(2b_1-a_1)\hat{\boldsymbol{\imath}}+(2b_2-a_2)\hat{\boldsymbol{\jmath}} = 11\hat{\boldsymbol{\imath}}+0\hat{\boldsymbol{\jmath}}$$

Equating the coefficients of $\hat{\boldsymbol{\imath}}$ and $\hat{\boldsymbol{\jmath}}$ gives:

$2b_1-a_1 = 11$ → Equation 1

$2b_2-a_2 = 0$ → Equation 2

Rearranging Equation 1 gives:

$a_1 = 2b_1-11$

Rearranging Equation 2 gives:

$a_2 = 2b_2$

Now,

$$|2\boldsymbol{b} - \boldsymbol{a}| = |11\hat{\boldsymbol{\imath}}|$$

$$|(2b_1 - a_1)\hat{\boldsymbol{\imath}} + (2b_2 - a_2)\hat{\boldsymbol{\jmath}}| = |11\hat{\boldsymbol{\imath}} + 0\hat{\boldsymbol{\jmath}}|$$

$$\sqrt{(2b_1 - a_1)^2 + (2b_2 - a_2)^2} = \sqrt{(11)^2 + (0)^2}$$

$$(2b_1 - a_1)^2 + (2b_2 - a_2)^2 = (11)^2$$

$$4(b_1)^2 - 4b_1a_1 + (a_1)^2 + 4(b_2)^2 - 4b_2a_2 + (a_2)^2 = 121$$

$$4[(b_1)^2 + (b_2)^2] + [(a_1)^2 + (a_2)^2] - 4(b_1a_1 + b_2a_2) = 121$$

$$4(100) + (169) - 4(b_1a_1 + b_2a_2) = 121$$

$$400 + 169 - 4(b_1a_1 + b_2a_2) = 121$$

$$569 - 4(b_1a_1 + b_2a_2) = 121$$

$$569 - 121 = 4(b_1a_1 + b_2a_2)$$

$$448 = 4(b_1a_1 + b_2a_2)$$

$$\frac{448}{4} = b_1a_1 + b_2a_2$$

$$112 = b_1a_1 + b_2a_2$$

Substituting $a_1 = 2b_1 - 11$ and $a_2 = 2b_2$ into $112 = b_1a_1 + b_2a_2$ gives:

$$112 = b_1(2b_1 - 11) + b_2(2b_2)$$

$$112 = 2(b_1)^2 - 11b_1 + 2(b_2)^2$$

$$112 = 2[(b_1)^2 + (b_2)^2] - 11b_1$$

$$112 = 2(100) - 11b_1$$

$$112 = 200 - 11b_1$$

$$11b_1 = 200 - 112$$

$$11b_1 = 88$$

$$b_1 = \frac{88}{11}$$

$$b_1 = 8$$

Substituting $b_1 = 8$ into $a_1 = 2b_1 - 11$ gives:

$a_1 = 2(8) - 11$

$= 16 - 11$

$= 5$

Substituting $a_1 = 5$ and $b_1 = 8$ into $112 = b_1a_1 + b_2a_2$ gives:

$112 = (8)(5) + b_2a_2$

$112 = 40 + b_2a_2$

$b_2a_2 = 112 - 40$

$b_2a_2 = 72$

Substituting $a_2 = 2b_2$ into $b_2a_2 = 72$ gives:

$b_2(2b_2) = 72$

$2(b_2)^2 = 72$

$(b_2)^2 = \frac{72}{2}$

$(b_2)^2 = 36$

$b_2 = \pm 6$

Substituting $b_2 = \pm 6$ into $b_2a_2 = 72$ gives:

$(\pm 6)a_2 = 72$

$a_2 = \pm\frac{72}{6}$

$a_2 = \pm 12$

$\therefore \boldsymbol{a} = 5\hat{\boldsymbol{\imath}} - 12\hat{\boldsymbol{\jmath}}$ and $\boldsymbol{b} = 8\hat{\boldsymbol{\imath}} - 12\hat{\boldsymbol{\jmath}}$ or $\boldsymbol{a} = 5\hat{\boldsymbol{\imath}} + 12\hat{\boldsymbol{\jmath}}$ and $\boldsymbol{b} = 8\hat{\boldsymbol{\imath}} + 12\hat{\boldsymbol{\jmath}}$

(b) The line L has equation $x - y + 1 = 0$ and the circle C has equation

$x^2 + y^2 - 2y - 15 = 0$.

(i) Show that L passes through the centre of C. [2]

$$x^2 + y^2 - 2y - 15 = 0$$

$$(x - 0)^2 + y^2 - 2y + 1 = 15 + 1$$

$$(x - 0)^2 + (y - 1)^2 = 16$$

$$(x - 0)^2 + (y - 1)^2 = 4^2$$

which is in the form $(x - a)^2 + (y - b)^2 = r^2$,

where the centre is $C(0, 1)$ and $r = 4$ units.

Substituting $x = 0$ and $y = 1$ into $x - y + 1$ gives:

$$(0) - (1) + 1 = -1 + 1$$

$$= 0$$

$$= \text{R.H.S.}$$

$\therefore$ The line L passes through the centre of C.

Q.E.D.

(ii) If L intersects C at P and Q, determine the coordinates of P and Q. [3]

$x^2 + y^2 - 2y - 15 = 0$ $\rightarrow$ Equation 1

$x - y + 1 = 0$ $\rightarrow$ Equation 2

Rearranging Equation 2 gives:

$y = x + 1$ $\rightarrow$ Equation 3

Substituting Equation 3 into Equation 1 gives:

$$x^2 + y^2 - 2y - 15 = 0$$

$$x^2 + (x+1)^2 - 2(x+1) - 15 = 0$$

$$x^2 + x^2 + 2x + 1 - 2x - 2 - 15 = 0$$

$$2x^2 - 16 = 0$$

$$x^2 - 8 = 0$$

$$(x + \sqrt{8})(x - \sqrt{8}) = 0$$

Either $x + \sqrt{8} = 0$ or $x - \sqrt{8} = 0$

$x = -\sqrt{8}$ $\quad$ $x = \sqrt{8}$

When $x = -\sqrt{8}$,

$$y = -\sqrt{8} + 1$$

$$= 1 - \sqrt{8}$$

When $x = \sqrt{8}$,

$$y = \sqrt{8} + 1$$

$$= 1 + \sqrt{8}$$

$\therefore$ The coordinates are $P(-\sqrt{8}, 1 - \sqrt{8})$ and $Q(\sqrt{8}, 1 + \sqrt{8})$.

(iii) Find the constants a, b and c such that $x = b + a\cos\theta$ and $y = c + a\sin\theta$ are parametric equations (in parameter θ) of C. [3]

$$x = b + a\cos\theta \qquad\qquad y = c + a\sin\theta$$

$$x - b = a\cos\theta \qquad\qquad y - c = a\sin\theta$$

$$\frac{x-b}{a} = \cos\theta \qquad\qquad \frac{y-c}{a} = \sin\theta$$

$$\left(\frac{x-b}{a}\right)^2 = \cos^2\theta \qquad\qquad \left(\frac{y-c}{a}\right)^2 = \sin^2\theta$$

Recall that,

$$\cos^2\theta + \sin^2\theta = 1$$

$$\left(\frac{x-b}{a}\right)^2 + \left(\frac{y-c}{a}\right)^2 = 1$$

$$\frac{(x-b)^2}{a^2} + \frac{(y-c)^2}{a^2} = 1$$

$$(x-b)^2 + (y-c)^2 = a^2$$

which is in the form $(x-0)^2 + (y-1)^2 = 4^2$,

where $b = 0, c = 1$ and $a = 4$.

$\therefore$ The values are $a = 4, b = 0$ and $c = 1$.

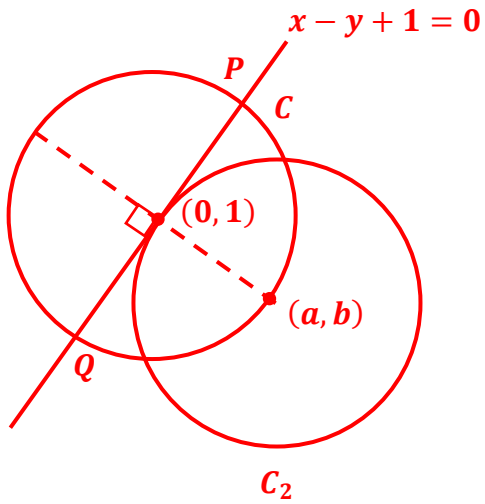
(iv) Another circle C_2 , with the same radius as C, touches L at the centre of C. Find the possible equations of C_2. [7]

Let the coordinates of the centre of C_2 be (a, b). The radius is 4 units.

Let the equation of C_2 be:

$$(x-a)^2 + (y-b)^2 = 4^2$$

Consider the diagram below:



Equation of line L is:

$x - y + 1 = 0$

$x + 1 = y$

$y = x + 1$

Gradient of line $L = 1$

Gradient of perpendicular line $= -1$

Consider points $(0, 1)$ and (a, b).

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$-1 = \frac{b-1}{a-0}$

$-1 = \frac{b-1}{a}$

$-a = b - 1$

$b = 1 - a$

The length between the points $(0, 1)$ and (a, b) is 4 units.

Length $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$4 = \sqrt{(a - 0)^2 + (b - 1)^2}$$

$$4 = \sqrt{a^2 + (b - 1)^2}$$

$$4^2 = a^2 + (b - 1)^2$$

$$16 = a^2 + b^2 - 2b + 1$$

$$16 - 1 = a^2 + b^2 - 2b$$

$$15 = a^2 + b^2 - 2b$$

Substituting $b = 1 - a$ into $15 = a^2 + b^2 - 2b$ gives:

$$15 = a^2 + (1 - a)^2 - 2(1 - a)$$

$$15 = a^2 + 1 - 2a + a^2 - 2 + 2a$$

$$15 = 2a^2 - 1$$

$$2a^2 = 15 + 1$$

$$2a^2 = 16$$

$$a^2 = \frac{16}{2}$$

$$a^2 = 8$$

$$a = \pm\sqrt{8}$$

When $a = \sqrt{8}$,

$$b = 1 - \sqrt{8}$$

When $a = -\sqrt{8}$

$$b = 1 - (-\sqrt{8})$$

$$= 1 + \sqrt{8}$$

$\therefore$ The equation of C_2 is:

$$\left(x-\sqrt{8}\right)^2+\left[y-\left(1-\sqrt{8}\right)\right]^2=4^2$$

or

$$\left(x+\sqrt{8}\right)^2+\left[y-\left(1+\sqrt{8}\right)\right]^2=4^2$$

Total: 25 marks

June 2011 – Question 4 – Solution

(a) By using $x = \cos^2\theta$, or otherwise, find all values of the angle θ such that

$$8\cos^4\theta - 10\cos^2\theta + 3 = 0,\ 0 \le \theta \le \pi. \qquad [6]$$

$$8\cos^4\theta - 10\cos^2\theta + 3 = 0$$

$$(2\cos^2\theta - 1)(4\cos^2\theta - 3) = 0$$

Either $\quad 2\cos^2\theta - 1 = 0 \quad$ or $\quad 4\cos^2\theta - 3 = 0$

$$2\cos^2\theta = 1 \qquad\qquad 4\cos^2\theta = 3$$

$$\cos^2\theta = \frac{1}{2} \qquad\qquad \cos^2\theta = \frac{3}{4}$$

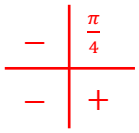
$$\cos\theta = \pm\frac{1}{\sqrt{2}} \qquad\qquad \cos\theta = \pm\frac{\sqrt{3}}{2}$$

Consider,

$$\cos\theta = \pm\frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}, \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

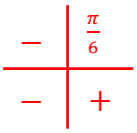


Consider,

$$\cos\theta = \pm\frac{\sqrt{3}}{2}$$

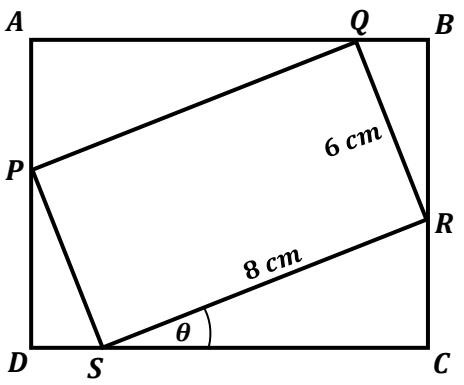
$$\theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$



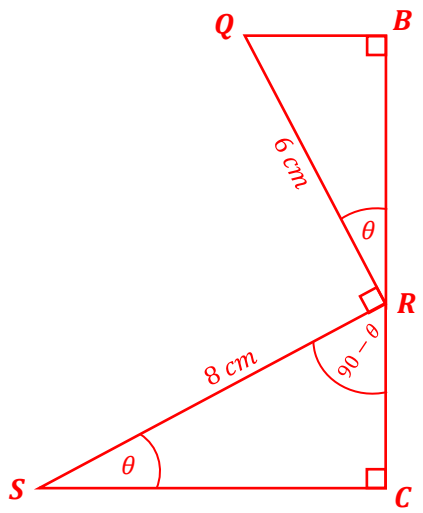
$$\therefore \theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{6} \text{ for } 0 \le \theta \le \pi.$$

(b) The diagram below, not drawn to scale, shows a rectangle $PQRS$ with sides 6 cm and 8 cm inscribed in another rectangle $ABCD$.



(i) The angle that SR makes with DC is θ. Find, in terms of θ, the length of the side BC. [2]

Consider the diagram below:



Consider the right-angled triangle SRC.

$\sin\theta = \frac{opp}{hyp}$

$\sin\theta = \frac{RC}{8}$

$RC = 8\sin\theta$

Consider the right-angled triangle QBR.

$\cos\theta = \frac{adj}{hyp}$

$\cos\theta = \frac{BR}{6}$

$BR = 6\cos\theta$

Now,

$BC = BR + RC$

$= 6\cos\theta + 8\sin\theta$

$\therefore$ The length of the side $BC = 6\cos\theta + 8\sin\theta$.

(ii) Find the value of θ if $|BC| = 7\ cm$. [5]

$|BC| = 6\cos\theta + 8\sin\theta$

Let us represent this expression in the form $R\cos(\theta - \alpha)$.

Let $6\cos\theta + 8\sin\theta = R\cos(\theta - \alpha)$

$6\cos\theta + 8\sin\theta = R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$

Equating coefficients of $\cos\theta$ and $\sin\theta$ gives:

$R\cos\alpha = 6$ $\rightarrow$ Equation 1

$R\sin\alpha = 8$ $\rightarrow$ Equation 2

Equation 2 $\div$ Equation 1 gives:

$$\frac{R\sin\alpha}{R\cos\alpha} = \frac{8}{6}$$

$$\tan\alpha = \frac{4}{3}$$

$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\alpha = 53.1° \quad \text{(to 1 decimal place)}$$

$$R = \sqrt{(6)^2 + (8)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

So, we have,

$$6\cos\theta + 8\sin\theta = 10\cos(\theta - 53.1°)$$

We are given that $|BC| = 7$.

Hence,

$$10\cos(\theta - 53.1°) = 7$$

$$\cos(\theta - 53.1°) = \frac{7}{10}$$

$$\theta - 53.1° = \cos^{-1}\left(\frac{7}{10}\right)$$

$$\theta - 53.1° = 45.6°$$

$$\theta = 45.6° + 53.1°$$

$$\theta = 98.7°$$

$\therefore$ The value of $\theta = 98.7°$.

(iii) Is 15 a possible value for $|BC|$? **Give a reason for your answer.** [2]

Suppose that $|BC| = 15$.

Then, we have,

$10\cos(\theta - 53.1°) = 15$

$\cos(\theta - 53.1°) = \frac{15}{10}$

$\cos(\theta - 53.1°) = 1.5 \quad \text{but } -1 \leq \cos\theta \leq 1.$

Hence, 15 is not a possible value for $|BC|$.

(c) (i) Show that $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$. [3]

Taking L.H.S:

$$\frac{1-\cos 2\theta}{\sin 2\theta} = \frac{1-(1-2\sin^2\theta)}{2\sin\theta\cos\theta}$$

$$= \frac{1-1+2\sin^2\theta}{2\sin\theta\cos\theta}$$

$$= \frac{2\sin^2\theta}{2\sin\theta\cos\theta}$$

$$= \frac{\sin\theta}{\cos\theta}$$

$$= \tan\theta$$

$$= \text{R.H.S.}$$

$$\therefore \frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$$

Q.E.D.

(ii) Hence, show that

(a) $\frac{1-\cos 4\theta}{\sin 4\theta} = \tan 2\theta$ [3]

From part (i), $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$.

Let $\theta = 2A$.

Then, $\frac{1-\cos 4A}{\sin 4A} = \tan 2A$.

Replacing A with θ gives:

$\frac{1-\cos 4\theta}{\sin 4\theta} = \tan 2\theta$

$\therefore \frac{1-\cos 4\theta}{\sin 4\theta} = \tan 2\theta$

Q.E.D.

(b) $\frac{1-\cos 6\theta}{\sin 6\theta} = \tan 3\theta$ [2]

From part (i), $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$.

Let $\theta = 3A$.

Then, $\frac{1-\cos 3A}{\sin 3A} = \tan 3A$.

Replacing A with θ gives:

$\frac{1-\cos 6\theta}{\sin 6\theta} = \tan 3\theta$

$\therefore \frac{1-\cos 6\theta}{\sin 6\theta} = \tan 3\theta$

Q.E.D.

(iii) Using the results in (c)(i) and (ii) above, evaluate

$$\sum_{r=1}^{n}(\tan r\theta \sin 2r\theta + \cos 2r\theta)$$

where n is a positive integer. [2]

Note that $\tan r\theta \sin 2r\theta = 1 - \cos 2r\theta$ for $i = 1, \dots n$.

This was deduced from the previous parts of the question.

Hence,

$$\sum_{r=1}^{n}(\tan r\theta \sin 2r\theta + \cos 2r\theta) = \sum_{r=1}^{n}(1 - \cos 2r\theta + \cos 2r\theta)$$

$$= \sum_{r=1}^{n} 1$$

$= n$, where n is a positive integer

Total: 25 marks

June 2011 – Question 5 – Solution

(a) Find $\lim_{x \to -2} \frac{x^2+5x+6}{x^2-x-6}$. [4]

$$\lim_{x \to -2} \frac{x^2+5x+6}{x^2-x-6} = \lim_{x \to -2} \frac{(x+3)(x+2)}{(x-3)(x+2)}$$

$$= \lim_{x \to -2} \frac{x+3}{x-3}$$

$$= \frac{(-2)+3}{(-2)-3}$$

$$= -\frac{1}{5}$$

$$\therefore \lim_{x \to -2} \frac{x^2+5x+6}{x^2-x-6} = -\frac{1}{5}$$

(b) The function f on $\mathbb{R}$ is defined by

$$f(x) = \begin{cases} x^2 + 1 & if\ x \geq 2 \\ bx + 1 & if\ x < 2 \end{cases}$$

Determine

(i) $f(2)$ [2]

When $x = 2, f(x) = x^2 + 1$.

$$f(2) = (2)^2 + 1$$

$$= 4 + 1$$

$$= 5$$

(ii) $\lim_{x \to 2^+} f(x)$ [2]

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (x^2 + 1)$$

$$= (2)^2 + 1$$

$$= 4 + 1$$

$$= 5$$

(iii) $\lim_{x \to 2^-} f(x)$ in terms of the constant b [2]

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (bx + 1)$$

$$= b(2) + 1$$

$$= 2b + 1$$

(iv) the value of b such that f is continuous at $x = 2$. [4]

Since f is continuous at $x = 2$, then

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$$

$$2b + 1 = 5$$

$$2b = 5 - 1$$

$$2b = 4$$

$$b = \frac{4}{2}$$

$$b = 2$$

$\therefore$ The value of $b = 2$.

(c) The curve $y = px^3 + qx^2 + 3x + 2$ passes though the point $T(1, 2)$ and its gradient at T is 7. The line $x = 1$ cuts the x-axis at M, and the normal to the curve at T cuts the x-axis at N.

Find

(i) the values of the constants p and q [6]

$y = px^3 + qx^2 + 3x + 2$

At $T(1, 2)$, $x = 1$ and $y = 2$.

Substituting $x = 1$ and $y = 2$ into the equation of the curve gives,

$2 = p(1)^3 + q(1)^2 + 3(1) + 2$

$2 = p + q + 3 + 2$

$p + q = -3$

Now,

$y = px^3 + qx^2 + 3x + 2$

$\frac{dy}{dx} = 3px^2 + 2qx + 3$

At $T(1,2)$, gradient at T is 7.

Substituting $\frac{dy}{dx} = 7$ and $x = 1$ gives,

$7 = 3p(1)^2 + 2q(1) + 3$

$7 = 3p + 2q + 3$

$3p + 2q = 4$

We have two simultaneous equations:

$p + q = -3$ → Equation 1

$3p + 2q = 4$ → Equation 2

Multiplying Equation 1 by 2 gives:

$2p + 2q = -6$ → Equation 3

Equation 2 − Equation 3 gives:

$p = 10$

Substituting $p = 10$ into Equation 1 gives:

$10 + q = -3$

$q = -3 - 10$

$q = -13$

The values of the constants p and q are $p = 10$ and $q = -13$.

$\therefore y = 10x^3 - 13x^2 + 3x + 2$

(ii) the equation of the normal to the curve at T [3]

Gradient of the tangent at $T = 7$

Gradient of the normal at $T = -\frac{1}{7}$

[since the product of the gradients of perpendicular lines $= -1$]

Substituting $m = -\frac{1}{7}$ and point $(1, 2)$ into $y - y_1 = m(x - x_1)$ gives:

$$y - 2 = -\frac{1}{7}(x - 1)$$

$$y = -\frac{1}{7}x + \frac{1}{7} + 2$$

$$y = -\frac{1}{7}x + \frac{15}{7}$$

$$7y = -x + 15$$

$\therefore$ The equation of the normal to the curve at T is: $7y = -x + 15$

(iii) the length of MN. [2]

The line $x = 1$ cuts the x-axis at M.

Therefore, the coordinates of $M = (1,0)$.

The normal to the curve at T cuts the x-axis at N.

When $y = 0$,

$$7(0) = -x + 15$$

$$x = 15$$

Therefore, the coordinates of $N = (15,0)$.

$$\text{Length of } MN = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(15 - 1)^2 + (0 - 0)^2}$$

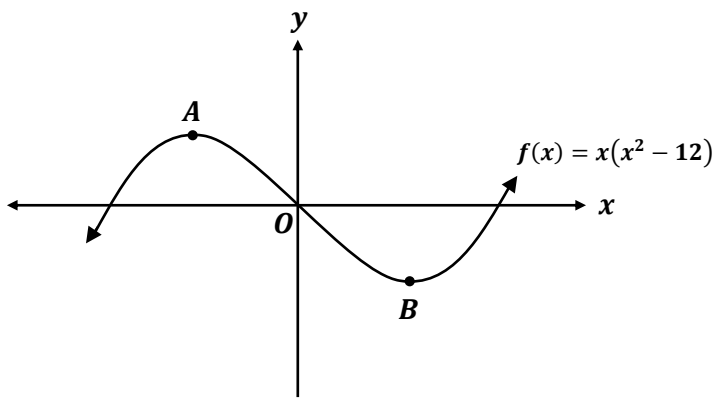
$$= \sqrt{(14)^2}$$

$$= 14 \text{ units}$$

Total: 25 marks

June 2011 – Question 6 – Solution

(a) The diagram below, **not drawn to scale**, is a sketch of the section of the function $f(x) = x(x^2 - 12)$ which passes through the origin O. A and B are stationary points on the curve.



Find

(i) the coordinates of each of the stationary points A and B [8]

$f(x) = x(x^2 - 12)$

$f(x) = x^3 - 12x$

$f'(x) = 3x^2 - 12$

At stationary points, $f'(x) = 0$.

When $f'(x) = 0$,

$3x^2 - 12 = 0$

$x^2 - 4 = 0$

$(x + 2)(x - 2) = 0$

Either $x + 2 = 0$ or $x - 2 = 0$

$x = -2$ $x = 2$

When $x = -2$,

$f(-2) = (-2)^3 - 12(-2)$

$= -8 + 24$

$= 16$

When $x = 2$,

$f(2) = (2)^3 - 12(2)$

$= 8 - 24$

$= -16$

$\therefore$ The coordinates are $A(-2, 16)$ and $B(2, -16)$.

(ii) the equation of the normal to the curve $f(x) = x(x^2 - 12)$ at the origin, O [2]

$f'(x) = 3x^2 - 12$

At the origin, $x = 0$.

When $x = 0$,

$f'(0) = 3(0)^2 - 12$

$= 0 - 12$

$= -12$

Gradient of tangent at $O = -12$

Gradient of normal at $O = \frac{1}{12}$

Substituting $m = \frac{1}{12}$ and point (0,0) into $y - y_1 = m(x - x_1)$ gives:

$y - 0 = \frac{1}{12}(x - 0)$

$y = \frac{1}{12}x$

$\therefore$ The equation of the normal is $y = \frac{1}{12}x$.

(iii) the area between the curve and the positive x-axis. [6]

$f(x) = x(x^2 - 12)$

$= x(x + \sqrt{12})(x - \sqrt{12})$

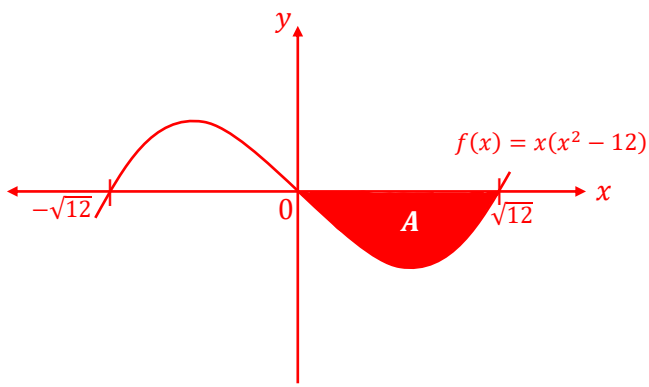
When $f(x) = 0$,

$x(x + \sqrt{12})(x - \sqrt{12}) = 0$

Either $x = 0$ or $x + \sqrt{12} = 0$ or $x - \sqrt{12} = 0$

$x = -\sqrt{12}$ $x = \sqrt{12}$

Consider the diagram below:



The area between the curve and the positive x-axis is labelled A.

$$A = \int_a^b f(x)\,dx$$

$$= \int_0^{\sqrt{12}} (x^3 - 12x)\,dx$$

$$= \left[\frac{x^4}{4} - 6x^2\right]_0^{\sqrt{12}}$$

$$= \left[\frac{\left(\sqrt{12}\right)^4}{4} - 6\left(\sqrt{12}\right)^2\right] - \left[\frac{(0)^4}{4} - 6(0)^2\right]$$

$$= \left[\frac{144}{4} - 6(12)\right] - \left[\frac{0}{4} - 6(0)\right]$$

$$= 36 - 72 - 0$$

$$= -36\ units^2$$

The negative sign indicates that the shaded region is below the x-axis.

So, we have,

$$A = |-36|$$

$$= 36\ units^2$$

$\therefore$ The area between the curve and the positive x-axis is $36\ units^2$.

(b) (i) Use the result

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx, a > 0\,,$$

to show that $\int_0^\pi x\sin x\ dx = \int_0^\pi (\pi - x)\sin x\ dx$. [2]

Let $f(x) = x\sin x$.

Let $a = \pi - x$

$\frac{da}{dx} = -1$

$dx = -da$

Also, $x = \pi - a$.

When $x = 0$,

$a = \pi - 0$

$a = \pi$

When $x = \pi$,

$a = \pi - \pi$

$a = 0$

So, we have

$$\int_0^\pi x\sin x\,dx = \int_0^\pi f(x)\,dx$$

$$= \int_\pi^0 f(\pi - a)\,(-1)\,da$$

$$= -\int_\pi^0 f(\pi - a)\,da$$

$$= -(-1)\int_0^\pi f(\pi - a)\,da$$

$$= \int_0^\pi f(\pi - a)\,da$$

$$= \int_0^\pi (\pi - a)\sin(\pi - a)\,da$$

Now,

$$\sin(\pi - a) = \sin\pi\cos a - \cos\pi\sin a$$

$$= (0)\cos a - (-1)\sin a$$

$$= 0 + \sin a$$

$$= \sin a$$

Hence,

$$\int_0^\pi x\sin x\,dx = \int_0^\pi (\pi - a)\sin(\pi - a)\,da$$

$$= \int_0^\pi (\pi - a)(\sin a)\,da$$

$$= \int_0^\pi (\pi - x)\sin x\;dx \qquad \text{since } \int_a^b f(x)\,dx = \int_a^b f(t)\,dt$$

$$\therefore \int_0^\pi x\sin x\;dx = \int_0^\pi (\pi - x)\sin x\;dx$$

Q.E.D.

(ii) Hence, show that

(a) $\int_0^\pi x\sin x\;dx = \pi\int_0^\pi \sin x\,dx - \int_0^\pi x\sin x\,dx$ [2]

Taking L.H.S:

$$\int_0^\pi x\sin x\,dx = \int_0^\pi (\pi - x)\sin x\,dx$$

$$= \int_0^\pi (\pi\sin x - x\sin x)\,dx$$

$$= \int_0^\pi \pi\sin x\,dx - \int_0^\pi x\sin x\,dx$$

$$= \pi\int_0^\pi \sin x\,dx - \int_0^\pi x\sin x\,dx$$

$$\therefore \int_0^\pi x\sin x\,dx = \pi\int_0^\pi \sin x\,dx - \int_0^\pi x\sin x\,dx$$

Q.E.D.

(b) $\int_0^{\pi} x \sin x \, dx = \pi$ [5]

$$\int_0^{\pi} x \sin x \, dx = \pi \int_0^{\pi} \sin x \, dx - \int_0^{\pi} x \sin x \, dx$$

$$\int_0^{\pi} x \sin x \, dx + \int_0^{\pi} x \sin x \, dx = \pi \int_0^{\pi} \sin x \, dx$$

$$2 \int_0^{\pi} x \sin x \, dx = \pi \int_0^{\pi} \sin x \, dx$$

$$2 \int_0^{\pi} x \sin x \, dx = \pi[-\cos x]_0^{\pi}$$

$$2 \int_0^{\pi} x \sin x \, dx = -\pi[\cos(\pi) - (\cos(0))]$$

$$2 \int_0^{\pi} x \sin x \, dx = -\pi[(-1) - (1)]$$

$$2 \int_0^{\pi} x \sin x \, dx = -\pi(-2)$$

$$2 \int_0^{\pi} x \sin x \, dx = 2\pi$$

$$\int_0^{\pi} x \sin x \, dx = \frac{2\pi}{2}$$

$$\int_0^{\pi} x \sin x \, dx = \pi$$

$$\therefore \int_0^{\pi} x \sin x \, dx = \pi$$

Q.E.D.

Total: 25 marks